



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 10.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MAC F314
Course Title : Mathematical Modelling
Instructor-in-Charge : DHIRAJ KUMAR DAS (6020 O, dhiraj.das@pilani.bits-pilani.ac.in)
Instructor(s) :
Tutorial/Practical Instructors: DHIRAJ KUMAR DAS

1. Course Description: Mathematical Modeling in Biology, Epidemiology, Model for Various Problems and Situation Using ODE and PDE, Classic Models in Ecology and Evolutionary Biology, Equilibria and Stability Analysis, Dynamics of Class-Structured Populations, Techniques for Analyzing Models.

2. Scope and Objective of the Course: The principal objective of this course is to familiarize students with the theory of population and a basic understanding of mathematical modelling of a real-world problem. Students will gain enough background to read theoretical population and evolutionary ecology literature, do simple modelling, and springboard to more complex theory if desired. Students will develop techniques for understanding and analysing simple, multivariate, and nonlinear models will be explored.

3. Course Learning Outcomes:

- CLO1.** Develop mathematical models of physical and biological systems using ODE.
- CLO2.** Understanding the concept of qualitative analysis of the model equations. Particularly, equilibria analysis, linear stability analysis and bifurcation theory.
- CLO3.** Analyze ecological and epidemiological models to interpret their dynamical behaviour and derive meaningful biological insights.
- CLO4.** Implement and simulate mathematical models using MATLAB and interpret the computational results..

4. Text Books:

- [TB1]. Mathematical Modeling with Case studies; B Barnes & G R Fulford, CRC Press, 3rd Edition, 2014.

5. Reference Books:

- [RB1]. Essential Mathematical Biology; Britton, Nicholas F, Springer, 2004.
- [RB2]. Mathematical Modeling: a tool for problem solving in engineering, physical, biological, and social sciences by D. N. P. Murthy, N. W. Page, Ervin Y. Rodin, Butterworth-heinemann Ltd, 1990
- [RB3]. Mathematical Biology I. An Introduction; James D. Murray, Springer-Verlag, 3rd Edition, 2005.



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6. Course Plan:

Module No.	Lecture Session	Reference from the Text Book	Learning outcomes
1.	L1.1-1.2:Mathematical Models and Modeling approaches	1.1-1.3	Introduction to Mathematical Models
2.	L2.1-2.3:Models related to Exponential decay, Lake Pollution, drug assimilation,	2.3, 2.5, 2.7	Understanding some physical system models
3.	L3.1-3.5:Exponential, Logistic, Density Dependent Growth, Limited Growth with Harvesting	3.1-3.3	Understanding the population growth models
4.	L4.1-4.4: Linearization and Critical Points	7.1-7.4	Understanding the concept of stability theory
5.	L5.1-5.6: Epidemic model for influenza, predator prey model and their phase plane analysis	5.2, 5.3, 6.1, 6.2, 6.4, R1: 2.3	Understanding the concept of models in ecology and epidemiology, their analysis, and drawing information
6.	L6.1-6.6: Population models of interaction species	R1: 2.4-2.6	Understanding the concept of modelling in biological species and their dynamical behavior.
7.	L7.1-7.8: SIS, SIR models, vector borne diseases, HIV infection model	R1:3.1- 3.5, 3.7, 6.5	Classical models in Epidemiology
8.	L8.1-8.6: Solving Models using MATLAB	----	Understanding the concept of computer simulation

7. Evaluation Scheme:

Component	Weightage [%] (Marks)	Duration	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	35% (70)	90 min.	As announced by AUGSD	Closed Book
Comprehensive Examination	45% (90)	3 h	As announced by AUGSD	Open Book
Quiz + seminar* (Continuous assessment)	20% (40)	20 min for each quiz; 10 min for seminar	To be announced by the instructor	Closed Book + Take-home



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* There will be three announced quizzes of 15 marks each, to be conducted during the tutorial hours. The best two quiz scores will be counted. The seminar presentation and submission of the seminar report will carry 10 marks. The schedule for the seminar presentations will be announced in due course.

Criterion for NC: (i) A total score of less than 10% of the maximum marks (i.e., less than 20 out of 200).
(ii) Absence in any one of the three evaluative components (i.e., Mid-semester Test, Comprehensive Exam, or entire Continuous Assessment).

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

9. Chamber Consultation Hour: Will be announced in the class.

10. Notices: All notices concerning this course will be displayed on the Nalanda website.

11. Make-up Policy: (i) In midsem and comprehensive exam, make-up will be given ONLY if the case is very genuine, and prior approval must be obtained from the IC.

(ii) **No makeup will be given for continuous assessment components.**

Instructor-in-charge
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